MGSC 672: Operations and Supply Chain Analytics

Individual Assignment

Submitted to:

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1 Foxconn & Flextronics Questions

1.1 What is the optimal order size and order frequency with each of Foxconn and Flextronics separately (12 points)?

We can determine the optimal inventory based on the economic order quantity or EOQ approach.

$$Q^* = \sqrt{\frac{2DS}{hC}}$$

D is annualized demand, S is the fixed cost incurred per order, h is the holding cost, c is the cost per unit, and Q* is the optimal ordering size.

We can determine the order frequency simply by applying the following calculation:

$$Order\ Freq = \frac{Annual\ Demand}{Q^*}$$

We will assume this question:

1. Demand is steady
2. No shortages are allowed and therefore inventory must be stratified
3. Replenishment time, otherwise known as lead time is fixed to zero in this case.

We will calculate the EOQ Foxconn and Flextronics separately.

Foxconn

$$Q^* = \sqrt{\frac{2 * 120{,}000 * 10{,}000}{0.25 * 100}} = \sqrt{\frac{2{,}400{,}000{,}000}{25}} = 9{,}797.95$$

$$Order\ Freq = \frac{120{,}000}{9{,}798} = 12.24$$

Flextronics

$$Q^* = \sqrt{\frac{2 * 48{,}000 * 10{,}000}{0.25 * 400}} = 3{,}098.38$$

$$Order\ Freq = \frac{48{,}000}{3{,}098} = 15.49$$

In other words, based on rounding void demand shortage, we should expect to order **9,798 smartphones/tables at a rate of 13 orders per year from Foxconn.** We should expect to order **3,098 laptops at a rate of 16 orders per year from Flextronics.**

1.2 For what fixed cost per order would an order size of 5,000 units be optimal for Foxconn (i.e, tablets/smartphones)? (6 points)

To calculate the fixed cost, we will simply need to rearrange the EOQ formula

$$Q^* = \sqrt{\frac{2DS}{hC}}$$

$$Q^{*2} = \frac{2DS}{hC} \Leftrightarrow S = \frac{hCQ^{*2}}{2D}$$

$$S = \frac{0.25 * 100 * 25{,}000{,}000}{2 * 120{,}000} = \frac{625{,}000{,}0000}{240{,}000} = 2{,}604.16$$

Therefore, the fixed cost per order should be **approximately $2,604.16.**

1.3 If the fixed cost of each shipment remains $10,000, what is the optimal order frequency and order size from the combined orders? How much reduction in cycle inventory can the company expect because of combining orders and shipments? (12 points)

To calculate the combined amounts, we will combine the demand and order frequency for both items.

Therefore for the EOQ calculation:

D=120,000+48,000

C= 120,000/(120,000+48,000)*100 + 48,000/(120,000+48,000)*400

=71.42+114.28

=185.71

$$Q^* = \sqrt{\frac{2DS}{hC}}$$

$$Q^* = \sqrt{\frac{2(120{,}000 + 48{,}000) * 10{,}000}{0.25 * (71.42 + 114.28)}} = \sqrt{\frac{3{,}360{,}000{,}000}{46.42}} = 8{,}507.01$$

$$Order\ Freq = \frac{168{,}000}{8{,}507} = 19.75$$

Therefore the **optimal order quantity for the combined orders is 8,507**, made up of approximately 6,076 smartphones and 2,431 laptops. The **optimal order frequency is 20 times** per year.

$$Cycle\ Inventory\ Combined = \frac{Q^*}{2} = \frac{8{,}507}{2} = 4{,}253.5$$

$$Cycle\ Inventory\ Separate = \frac{Q^*}{2} = \frac{9{,}797.95 + 3{,}098}{2} = 6{,}443$$

Therefore, the **cycle reduction for combining shipping is 2,190.**

2 Home Depot

2.1 Why can a Home Depot with a few large stores provide a higher level of product availability with lower inventories than a hardware store chain such as Tru-Value, with many small stores? (3 points)

Home Depot is a national home improvement and retail store that supplies building materials, equipment and does rentals. Home Depot is likely to prioritize increasing supply chain speed and accuracy of information over other retailers and therefore have a higher material availability.

The OM triangle is a useful visual tool to understand this. In a supply chain , a company can prioritize the Inventory, time, speed, or information of this supply chain. A company that focuses on inventory will stockpile inventory at a high cost, while companies like Zara prioritize the speed of their supply chain. Home Depot's operating model indicates they likely specialize in both speed and supply chain information to get the right amount of material at the right time with few mistakes.

Home Depot is becoming a data-powered organization that can access large amounts of data to reduce the uncertainty in their demand. Home Depot is increasing its investment in data to increase the speed and accuracy of its predictions for inventory stocking. [1] Consumers are also benefiting as the inventory for any item, and its location are readily available online. This helps consumers self-select and go to the store if the items are in stock. Furthermore, Home Depot has created new data collection channels like credit cards, customer loyalty, and customer profiles for as many customers as possible.

Home Depot has also made significant efforts to speed up its supply chain by investing in new fulfilment centers and better supply chain reorder automation.[2] By investing in supply relationships and inventory speed management Home Depot increases the effectiveness of its supply chain.

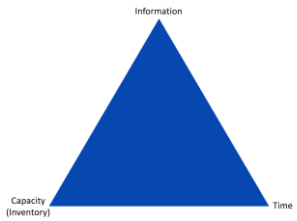


Figure 1: OM Triangle. Visualization of choice of prioritization that companies have to make about their supply chains.



Figure 2: OM Triangle with Example Companies

2.2 In the 1980s, paint was sold by color and size in paint retail stores. Today paint is mixed at the paint store according to the color desired. Discuss what, if any, impact this change has on safety inventories in the supply chain. (3 points)

This change should profoundly impact increasing the speed and reducing the complexity of the supply chain while decreasing the need for safety stock.

In the 1980s, paint stores may have only held inventory and stocked up on the most popular colours to satisfy the immediate demand of commercial companies and consumers. The remaining customers who wanted a custom colour would likely have to wait and use an order system. Many custom orders would tax the supply chain with, while unsold inventory would be marked down or disposed of, increasing waste and reducing revenues. Therefore, stores in the 80s would likely have had to keep large inventories of safety stock for unpredictable demand seasonality or product colours could be met.

Today the supply chain of paint focuses on shipping only a handful of base layers that could have colours added or even go on as layers themselves. The demand for one type of product is more predictable and stable, allowing better forecasting. Any large spikes, such as contractors could be processed through a custom order. More accurate forecasting, specializing bulk orders, and mixing in-store has likely dramatically decreased the overall safety stock within stores.

2.3 Any leftover units of one product are worthless. Leftover units of the other product can be sold to outlet stores. Which product should have a higher level of availability? Why? (3 points)

Items with a salvage value that can be sold back to stores should have higher availability. Any item that has a value if not sold is inherently less risky from a business perspective than those that cannot be resold. In other words, the cost of overstocking (Co) is much lower for items with salvage value and therefore, more of these items can be stocked while maintaining the same risk-reducing waste (as excess units will sell at outlets), and these items will increase revenue through additional volume of sales.

2.4 A firm improves its forecast accuracy using better market intelligence. What impact will this have on supply chain inventories and profitability? Why? (3 points)

First, accuracy in demand forecasting will decrease expected overstock, thus reducing waste. If companies have a better sense of how much demand to expect they will order or stock less and therefore be much closer to matching inventory to demand.

Second, the increased accuracy will decrease the expected understock of units. Suppose the company knows the maximum possible demand 95% of the time. In that case, they can accurately capture demand 95% of the time and thus reduce understock where demand exceeds the supply and the company loses potential sales.

These effects should culminate in increasing the expected profit of the organization while reducing waste.

3 Green Thumb

3.1 How many units should GT manufacture for sale? (8 points)

GT should stock 78 units for the season.

GT should balance the risk of over-ordering and the risk of under-ordering. We can balance these risks, often referred to as the cost of overstocking (Co) and the cost of understocking (Cu).

$$F(O) = \frac{Cu}{(Co + Cu)} = \frac{50}{(150 - 50 + 20 + 50)} = \frac{50}{170} = 0.29$$

Cu= Price of product- cost of product = 50

Cu= Cost of product – Salvage Value + Cost of inventory storage=120

Where F(O) is the distribution function of demand, otherwise known as the critical ratio.

We can use the critical ratio and the normal distribution to determine the order number. Using Excel logic we can use the following function NORMINV(0.29,100,40) to obtain the number of units we should order. This result gives 78.34 or rounded 78 units.

3.2 What is the expected profit? (3 points)

The expected profit with 78 units that are all sold throughout the season is $2,657. This value was calculated using the "Session 4B Safety Inventory Students" excel document. The formula here takes the price minus the salvage value to estimate the average revenue. Then the formula evaluates each other possibilities of overstock, understock, or perfect sell out using the *NORMDIST* function on the average demand.

$$\text{Expected profit} = (p - s)\mu F_s\left(\frac{O - \mu}{\sigma}\right) - (p - s)\sigma f_s\left(\frac{O - \mu}{\sigma}\right) - O(c - s)F(O, \mu, \sigma) + O(p - c)[1 - F(O, \mu, \sigma)]$$

Figure 3: Equation 13.3 expected profit.[3]

3.3 On average, how many units will be leftover at the end of the season and how many stockouts will be there? (4 points)

There will be an **expected 31 units of overstock** through the season based on equation 13.4. Furthermore, there will be an expected **56 units of understock** through the season. These numbers are based on the provided excel sheet and equations from Hofmann, Erik. "Supply Chain Management: Strategy, Planning and Operation, S. Chopra, P. Meindl." (2013).

$$\text{Expected overstock} = (O - \mu)F_S\left(\frac{O - \mu}{\sigma}\right) + \sigma f_S\left(\frac{O - \mu}{\sigma}\right)$$

Figure 4: Equation 13.4: Expected Overstock units[3]

$$\text{Expected understock} = (\mu - O)\left[1 - F_S\left(\frac{O-\mu}{\sigma}\right)\right] + \sigma f_S\left(\frac{O-\mu}{\sigma}\right)$$

Figure 5: Equation 13.5: Expected Understock Units[3]

3.4 How will the result for part a change if it costs GT $50 to dispose of the leftovers at the end of the season (every other thing remains the same)? (8 points)

If the units are not able to be sold at a post-season sale then the salvage value s becomes $-70 (a cost of 50 for disposal and 20 for stocking). Therefore the all the values of optimal inventory including order quantity and expected profit will change. The following table summarizes the impact.

	Cost of $50 for disposal of equipment	From question above ($50 salvage value from end of season)	Difference
Median Demand	100	100	
Standard deviation of Demand	40	40	
Cost per unit	$150	$150	
Price per unit	$200	$200	
Salvage value	$-70	$30	$100
Co	$220	$220	
Cu	$50	$50	
CSL	0.19	0.29	-.1
Order Size	64	78	-14
Expected profit	$2,520	$2,657	$-137
Expected overstock units	22	31	-9
Expected understock units	60	56	4

4 Walmart

4.1 Based on the case about Walmart.com included in your case pack what do you think are the different supply chain levers (inventory, warehousing, transportation, information, ..) that used by Walmart to effectively compete with Amazon? (35 points – 2 pages max)

Walmart has existed since the 1940's in some form. Their continued growth an investment over the past 80 years have made them a strong competitor. Several factors play a key role in their success despite tough competition, particularly in the ecommerce space. Walmart has use several supply chain features and levers effectively to compete with amazon.

First, Walmart made an investment in their transportation assets. Walmart owns their own trucks and delivery vehicles for their pallets and goods. Walmart purchased the assets and started to male this investment back om the 1980's. During this period, they became experts at cross docking, which enabled them to lower transportation supply chain costs and make the right decision about transportation optimization for their needs. As their business grew and w=Walmart developed more super stores, which carried food and groceries Walmart added many chilled trucks and additional capacity to their delivery networks. The ownership of the delivery vehicles and medium of transportation has helped them reduce cost and reduce uncertainty in their supply chain. In fact Amazon has been attempting to start and develop its delivery business. Amazon owns trucks, vans and planes for various types of shipping. Nevertheless, Walmart has the supply chain ownership already developed. Walmart's control of the supply chain is an effective competitive element against amazon.

Second, Walmart is controlling the content of their supply chain much more closely than that of Amazon. This is an advantage in the supply chain as fewer SKUs are much easier to manage than massive amounts of them. This is due to the cost efficiency of pallets, and volume discounts versus one off items. Amazon democratizes their product offering through their marketplace. The article mentions how BCG compared 20 Walmart products versus over 12,000 for Amazon. While this may appear to be an advantage for Amazon, in fact it's a practical choice for Walmart. Walmart can better control its brand and the quality of items. This means that any e-commerce Walmart, they can much more easily avoid and process returns. Returns have become a significant issue in the supply chain, some say a "trillion dollar problem"; amazon has often found it cheaper to destroy goods than process them.[4, 5] Returns are a significant problem for companies from a monetary, environmental and public, perception perspective. [5] By focusing on fewer essential products, Walmart has a competitive advantage. Walmart will also avoid the analysis paralysis trap more easily than Amazon, where many consumers have an overwhelming number of choices.

Third, Walmart uses their existing infrastructure to their advantage by buying online pick up in-store (BOPIS). Leveraging existing stores to encourage customers to pick up in-store or BOPIS enables a cost saving on the last mile delivery, using existing infrastructure and a lift effect on other stores. Last mile delivery could be more efficient, and quite expensive for companies. It is a complex activity for companies and needs to be adequately addressed. Many have sought to optimize routes through travelling salesman algorithms or reduce delivery via automated drones. Still, the high cost of delivery persists. Walmart is helping to avoid this issue by offering a free BOPIS pathway for the customer. This process is cost effective compared to other methods. BOPIS is an effective method of competing against Amazon to the point that Amazon has been installing lockers to help recover the cost of last mile delivery. Walmart can use existing infrastructure, such as their feel of trucks or associates, to help deliver the e-commerce strategy. Walmart has a clear competitive advantage since the infrastructure already exists and the marginal cost is low for BOPIS. By using their existing infrastructure to encourage customers to pick up in-store, Walmart has found that this generates a lift in-store sales by $30. Walmart is effectively using their supply chain by enabling customers to use its existing infrastructure and avoid the last mile delivery, which is generally the costliest portion of e-commerce.

Lastly, Walmart has invested in the speed of its supply chain and information. Walmart has been able to provide same-day or next-day pickup for many customers. The passage indicates that "the combined logistics system also allowed Walmart to serve 87% of the country overnight and 99% in 2 days. They are showing that the speed of their supply chain is highly efficient. Walmart invests in satellite networks, efficient supplier coordination and communication. A speedy supply chain can respond quickly to changes and adapt quickly. Walmart has also become a highly able information and data-driven organization. Suppliers like P&G share data and have access to the information to reduce the bullwhip effect and further increase forecast accuracy. Despite data sharing, Walmart has acquired many

companies that have pioneered or innovated e-commerce or data strategies. For instance, Walmart acquired Jet.com to augment its e-commerce platform and to get mark lore, who had previously worked with Amazon. Walmart can analyze millions of transactions per second and have high predictability levels in their demand. This means Walmart has enough resources to predict consumer behaviour and accurately deliver product needs.

Walmart uses the five levers mentioned above to compete effectively against the tech giant Amazon. Walmart uses their transportation network, limited quality products, BOPIS, speed and information within its supply chain to gain a competitive advantage over Amazon.

5 References

(1) Group, B. B. C. The Home Depot Doubles Down on Data Science. *Wired* **2023**.
(2) Goel, S. Business Model & Supply Chain Analysis of Home Depot. The Strategy Story: 2023.
(3) Chopra, S.; Meindl, P. Strategy, planning, and operation. *Supply Chain Management* **2001**, 13-17.
(4) Geiger, I. Amazon continues to destroy new goods and returns. *Pressenza*, 2022.
(5) Timlin, A. The high cost of e-commerce returns: A trillion dollar problem. *The Future of Commerce*, 2023.